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An Efficient Image Contrast Enhancement using Segmentation in Genetic Algorithm for Remote Sensing Applications

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***Abstract*—This paper proposes a novel optimized histogram equalization technique using a genetic algorithm (GA) for brightness-preserved contrast enhancement. This method optimizes the histogram equalization technique to extract detailed data from distorted images. The principal objective is to enhance histogram equalization by fixing a trade-off between contrast improvement and mean-brightness preservation. A genetic algorithm is used to generate the new target histogram. The optimized histogram is used in the histogram equalization (HE) process to generate the final enhanced image. The performance of the proposed method is checked and validated with qualitative and quantitative assessments against well-known state-of-the-art techniques. The result section depicts that the proposed method significantly enhances the visual effect of distorted images based on human subjective and objective evaluation indices. 3-5 percent enhancement can be observed if we compare the standard Histogram Equalization method to the proposed method.**

***Index Terms*—Genetic algorithm, Contrast enhancement, Optimization techniques, Histogram equalization**

I. Introduction

Information around us is present in many forms, and one such form is image pixels. These images contain information about the geological background, underwater information, or some technical information such as QR codes and Bar codes. Remote image processing is another technique through which we gather information about geological nature by sensing the image from a considerable distance (such as satellite images). While capturing such images from such a significant distance, there is a high probability that image information density, contrast, and saturation are not up to the mark due to weather conditions, device malfunctioning, or even errors in directional coordinates. This type of fault results in reducing the image information and clarity. Satellite images are one of the most informative images whenever we talk about the behavior of the earth and weather around the globe. Even minor errors in reading can have a vast effect on overall calculations for future use and predictions. Many techniques were introduced to overcome the challenges faced by less-contrasted less-informative images [1][2]. These techniques focus on improving the image's overall structure so that it is not entirely changed, and the content becomes more accurate and readable without losing its natural structure.

Using the Histogram Equalization (HE) technique, we can generate a uniform histogram output. It is one of the most effective methodologies to solve contract enhancement problems by overstretching the intensity level information to the available dynamic range [3]. We flatter the probability distribution function or PDF in the HE approach and increase the grey levels range. The cumulative density function(CDF) is derived from the probability distribution function to map the intensity regions of the original image. Though HE is an adaptive and precise technique for contrast enhancement, it may result in over-enhancement and noisy patches when used without modification [4][5].

In addition, HE updates the image's brightness to the middle of the gray level, which may also result in annoying artifacts in the output image. From this, we can conclude that HE is not applicable in every situation, especially where brightness maintenance is essential. Further, Brightness preservation is necessary to avoid annoying artifacts [6][5]; therefore, to overcome this issue, an efficient contrast enhancement technique based on a genetic algorithm (GA) is proposed in this paper. The proposed method uses a chromosome representation and related operators using the segment analysis method.

The paper has been divided into sections containing Related Work, Algorithm Analysis followed by Experimental Results. In the related work section, all the methods used and their significance have been discussed. Later, it is followed by the Proposed method section, which contains the main algorithm with mathematical equations, functions, and pre-processing. All these functions were run on the images, whose results have been included in the Experimental results section (both table and output images).

II. Related Work

The section contains the relevant schemes, information, and the working principles of the Histogram equalization algorithm. It also includes the working principle of the genetic algorithm that is used for contrast enhancement in the proposed section. The brightness preservation has been a measured challenge in related scientific contributions. Keeping this in view, Several Algorithms have been proposed to overcome the brightness preservation problem, mainly based on Histogram Equalization. The brightness-conserving Bi-HE (BBHE) approach maintains the mean brightness while expanding the image's contrast [7][8]. The other similar

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HE method is RMSHE(recursive mean dispersed histogram equalization), which produces better contrast enhancement which results in brightness protection [9]. Another method introduced by Wanetal, DSIHE (dualistic sub-image histogram equalization) claims better execution than BBHE in saving the information and brightness of the image. The difference is that this method separates the histogram for the median value rather than the average value. The splitting of the histogram has also been considered by another methodology introduced in 2007 named MWCVMHE (minimum within-class variance multi-HE) trailed by their respective histogram equalization of the sub-images. However, this method experiences a bit less optimized enhancement level [10][11].

Further, Abdullah et al. proposed a Dynamic histogram equalization(DHE) [12] which partitions the histogram into two sections, focusing on minor data. It is followed by the dynamic reach to every one of the sub-divided histograms that use the general HE interaction [13]. This cycle alters the brightness and means level of the image carrying the mapping of the highest bins of the histogram. The other alternative method for contrast enhancement, called BHEPL (bi-HE with plateau limit) [14], has been introduced to improve the original histogram equalization and conserves the originality of the image. Using this method, the histogram is initially partitioned into two segments given the average data of the image, trailed using the sub-histograms cutting with a limit. Image enhancement is generally accomplished on the sub-histograms using the histogram equalization technique. Intensity irregular distribution can be reduced by using a histogram level limit. To overcome this issue, ACMHE (Adaptive contrast enhancement using modified HE) [15][16][17] was introduced to isolate the histogram into four sections for its median value. The separation process is applied with the help of the mean in the image on every sub-histogram.

Another method of Histogram Equalization [16] using variance and standard deviation has been introduced, which is helpful for image contrast enhancement. The histogram equalization can be implemented independently. The contrast enhancement rate in this method is constrained by characterizing an argument [18]. All these methods are effective in contrast enhancement, but the major problem with these algorithms is that they produce images that are ineffective in keeping the mean intensity and do not look natural. Therefore these techniques are less helpful in extracting information from satellite images. To overcome this problem, a new technique has been proposed based on a genetic algorithm with segment separation [6][5]. The following section discusses the genetic algorithm in detail.

III. The Proposed Method

This section contains the working of the algorithm by describing the pre-processing functions, fitness function designed for the genetic algorithm, utilization of the new histogram and mapping the grey values from the previous histogram to the new histogram for generation of the enhanced image. The working of the algorithm has also been described with diagrams.

A. Genetic Algorithm

A genetic algorithm (GA) is a search heuristic inspired by Charles Darwin's theory of natural selection. This algorithm represents a population of individual solutions (called chromosomes, the genotype, or the phenotype) as strings [19]. Each chromosome can be seen as a unique solution to a problem. The fitness function is used to evaluate the obtained solution. The better a solution is, the higher its fitness value. The goal of a GA is to find the optimal solution to a problem by using a procedure called selection. Selection involves choosing the best individual solutions (chromosomes) from the current population and breeding them to produce the next generation. The new generation of solutions is then evaluated using the fitness function to see if any improvements have been achieved. If the new generation is better than the previous one, it becomes the current population. This process is repeated until the desired goal is reached. The main advantage of GAs is that they can be applied to problems that are difficult to solve using traditional methods. They can also be used to find solutions to problems with multiple objectives. The next section proposes a new efficient method using the knowledge and working principle of genetic algorithm.

B. Pre-processing Algorithm Analysis

The working principle of the proposed method is primarily based on a genetic algorithm that is optimized using a fitness function, which will be discussed in section 3.2 of the paper. The very first step of the proposed method includes the conversion of the 3-D color image matrix to 2-D grayscale image matrix which will be used as a input for the proposed genetic algorithm method. To convert the 3-D RGB matrix into a 2-D grayscale matrix, the following algorithmic formula is used:

$$Grayvalue(Gr) = 0.2989 * R + 0.5870 * G + 0.1140 * B \quad (1)$$

where Gr is the corresponding gray value of the Red, Green and Blue shades represented by R, G and B. All these Gr, R, G, and B will have a standard range of matrix image colours from 0 to 255 (both inclusive). Further, a histogram of this newly constructed grayscale matrix is obtained by taking the unique gray values. In simpler words, a 2-D image matrix is represented by a 1-D Histogram of gray levels which is further passed to the proposed genetic algorithm. The histogram is one of the effective ways to represent the image matrix in a one-dimensional analysis. For this purpose, the following algorithm is used to obtain the histogram from a 2-D image grayscale matrix.

C. Fitness Function for Genetic Algorithm

After obtaining the histogram of the image matrix, the next step is to obtain the new generations (enhanced values of the original histogram) of the histogram. In order to obtain these histograms the fitness function is required.

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Algorithm 1 Converting an image matrix into histogram

Require: $imageMat(imagematrix)$
Ensure: $A\,one-dimensional\,array\,hist\,is\,returned$
  $hist[256] \leftarrow [0]$
  $r \leftarrow rows(imageMat)$
  $c \leftarrow columns(imageMat)$
  for $i \leftarrow 0\,to\,r$ **do**
    for $j \leftarrow 0\,to\,c$ **do**
      $gray = imageMat[i][j]$
      $hist[gray] = hist[gray] + 1$
    end for
  end for

The primary significance of this fitness function is to determine the fitness of a histogram based on mapped values with the original histogram. In simple words, mapping of the original image matrix values is done to the new image matrix values. The fitness value of an image matrix will be calculated on the basis of two things - The number of edges of the new matrix and the relationship between the corresponding image pixels with their adjacent image pixels. This fitness function will generate a fitness score using the following formula:

$$fitness(X) = log(log(E(I(X)))) * edges(I(X)) \quad (2)$$

$$E(I(X)) = \Sigma_x \Sigma_y \sqrt{(h_1(x,y)^2 + v_1(x,y)^2)} \quad (3)$$

where edges(I(X)) represent the number of visible edges that can be calculated using the Edge Detector and two functions $h_1()$ and $v_1()$ will provide the measure of the dependency with respect to the current pixel (x, y) with its adjacent pixels. These functions are calculated as follows:

$$h_1(x,y) = P(x+1,y-1) + 2*P(x+1,y) + P(x+1,y+1) + P(x-1,y-1) - 2*P(x-1,y) - P(x-1,y+1) \quad (4)$$

$$v_1(x,y) = P(x-1,y+1) + 2*P(x,y+1) + P(x+1,y+1) + P(x-1,y-1) - 2*P(x,y-1) - P(x+1,y-1) \quad (5)$$

where P(x,y) refers to the gray value of the pixel at the position (x,y) in the image matrix.

D. Optimized Histogram using the Genetic Method

Now comes the major part of the algorithm, which is to use the genetic algorithm to find the best possible histogram in contrast enhancement. The initial histogram will be represented by $hist$ as a variable name. The new histogram will keep on generating using the get_chromosome() method which is not generating a random histogram but is based on the properties of equalization and segments. As discussed earlier, histogram equalization is an image processing technique that enhances contrast in images. It spreads out the most frequent intensity values, enhancing the image's contrast. The idea of generating the best possible histogram is to ensure that the two randomly generated values are not very close. For example, if the algorithm has generated a value of 20, then it should make sure that the next generated value should be at least 30 or whatever based on the histogram available size. This leads to generating a value in segments rather than being random. A sample run on the histogram of size 5 with the two methods - Random and Segments is as follows:

Random Generation:
1. Get a random value between 0-255 = 34
2. Get a random value between 0-255 = 45
3. Get a random value between 0-255 = 100
4. Get a random value between 0-255 = 36
5. Get a random value between 0-255 = 56

Random Generation in Segments:
1. Get a random value in the segment 0-51 = 35
2. Get a random value in the segment 52-102 = 73
3. Get a random value in the segment 102-153 = 119
4. Get a random value in the segment 154-205 = 191
5. Get a random value in the segment 206-255 = 223

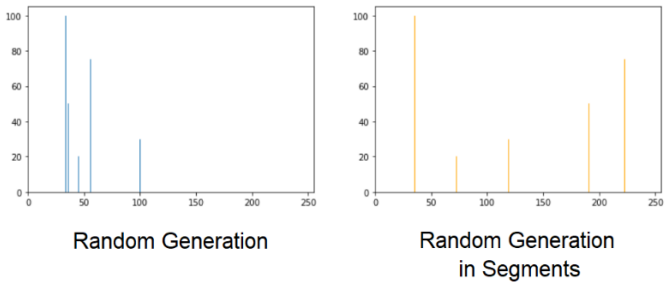


Fig. 1: Random Generation and Random Generation in Segments graph

It can be clearly seen from the graphs (Fig 1) that the histogram is better equalized in the right image as compared to the left which is because the values are generated in the different segments[14]. These ranges were not overlapping. Whereas in the first case, all the random values generated do not have a threshold gap in between so they are closer and the histogram is not properly equalized. This histogram equalization using segments is done using the following algorithm: Also, the number of iterations for this genetic algorithm should be limited because it may be possible that the algorithm will going to run an infinite number of times. The iterations are generally fixed in these types of randomized methods to avoid run-time errors. Further, this methodology has been explained through the flow chart (Fig. **??**). The genetic algorithm and its iterations are based on the following algorithm:

E. Mapping the Gray values to new Histogram

Now comes the part of mapping the results from the old histogram to the new histogram. The mapping is a simple one-to-one mapping done on the histograms. For example, if the original histogram is $[x_1, x_2, x_3...]$ and the new histogram

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Algorithm 2 Generating a new chromosome

Require: $hist(imagehistogram)$
Ensure: $A\,one-dimensional\,array\,newHist\,is\,returned$
$newHist \leftarrow []$
$N \leftarrow size(hist)$
$step \leftarrow max(255/N, 1)$
$lowerBound \leftarrow 0$
$upperBound \leftarrow step$
for $i \leftarrow 0\,to\,N$ **do**
$num \leftarrow random(lowerBound,\ upperBound)$
$newHist.push(num)$
$lowerBound \leftarrow upperBound$
$upperBound \leftarrow min(255, upperBound + (2 * step))$
end for

Algorithm 3 Genetic algorithm design

Require: $An\,image$ **img**
Ensure: $An\,Enhanced\,image\,matrix$ **newImgMat**
$result \leftarrow []$
$newHist \leftarrow []$
$imageMat \leftarrow getImageMat(img)$ ▷ eq. 1
$hist \leftarrow getHist(imageMat)$ ▷ From Algorithm 1
$maxIter \leftarrow 100$ ▷ Set Maximum Iterations
for $i \leftarrow 0\,to\,maxIter$ **do**
$newHist \leftarrow getNewChromosome(hist)$ ▷ From Algorithm 2
if $fitness(newHist) \geq fitness(hist)$ **then** ▷ Fitness eq. 2, eq. 3
$result \leftarrow newHist$
end if
end for
$newImgMat \leftarrow getNewImageMat(imgMat, hist, result)$ ▷ Algorithm 4

is $[y_1, y_2, y_3...]$ then all the pixels having x_1 as gray value are changed to y_1, x_2 gray values changed to y_2 and so on. This will be more clear from the following illustration (Fig. 2, Fig. 3): Mapping of the histogram values from the original one to

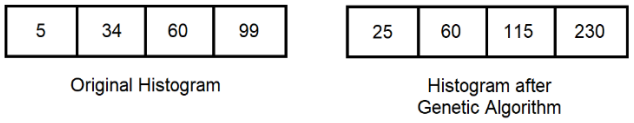


Fig. 2: Histogram generated on right-hand side from the left side using genetic algorithm (These values are arbitrary and not to be taken as original values)

the new one is based on the following algorithm:

IV. Experimental Results

In the field of image processing, contrast enhancement is one of the pioneering techniques. It has accumulated a variety of applications in significantly important domains like satellite image analysis etc. Remote sensing data is very much sensitive

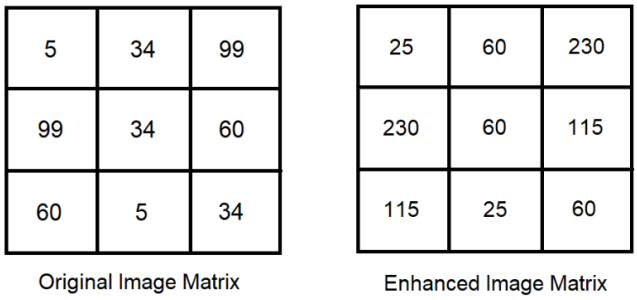


Fig. 3: Original Image matrix grayscale values are directly mapped to enhanced image matrix values with reference to Fig 3 Histograms

Algorithm 4 New image matrix using Histogram Mapping

Require: $Initial\,image$ **imgMat**, $Initial\,Histogram$ **hist**, Final Histogram, **result**
Ensure: $An\,Enhanced\,image\,matrix$ **newImgMat**
$newImgMat \leftarrow imgMat$
$histMap \leftarrow new\,map()$
$N \leftarrow size(hist)$
for $i \leftarrow 0\,to\,N$ **do**
$histMap[hist[i]] \leftarrow newHist[i]$
end for
$r \leftarrow rows(imgMat)$
$c \leftarrow columns(imgMat)$
for $i \leftarrow 0\,to\,r$ **do**
for $j \leftarrow 0\,to\,c$ **do**
$newGrayVal \leftarrow histMap[imgMat[i][j]]$
$newImgMat[i][j] \leftarrow newGrayVal$
end for
end for

and requires special attention while increasing the information contained in it. Keeping this in view, the newly proposed technique which has already been discussed in the previous section is tested here for its suitability.

The results are shown below on the various kind of images taken into consideration for comparison by different quality parameters. This comparison has been established between the standard contrast enhancement technique such that Histogram Equalization Image Enhancement Method (HE) and the proposed Genetic Algorithm based contrast enhancement technique with a segmented approach. The proposed algorithm is practically used on the images (shown in Fig. 6 - Fig. 13). The parameters that are included in the comparison of these methods are Mean, Entropy, Variance, and PSNR[1]. The closer the mean is to a middle value (between 110-130) the more likely the image has better contrast and is not too light or too dark[1].

On the other hand, variance decides the trade-off between the mean pixel values and the original pixel values, so if the variance is higher, then there are relatively more changes in the image compared to the original images. Further, Entropy decides the amount of information that the image contains. In

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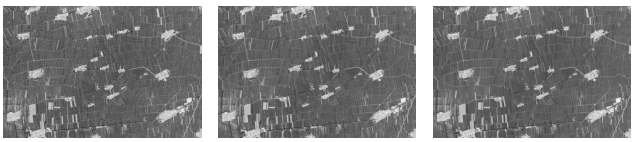
addition, the Entropy is directly proportional to the mean and variance. Conclusively, if there is no major change in mean and variance, there will not be a significant increase in the information[1]. This observation can be seen in the figures also.

Every figure (Fig. 5 to Fig. 12) is divided into three sub-figures: The first sub-part (Fig. 5(a)) of the image contains the original image that is tested with different contrast enhancement techniques, i.e., HE, Proposed GA technique. The second image (Fig.5(b)) is generated using the HE technique. In continuation, the third sub-part of the figure(Fig.5(c))) is the image generated using the proposed method based on the segmented GA concept. These figures depict that while applying the histogram equalization techniques to the original images, the visual effects are not as smooth as the proposed segmented GA technique. The proposed method drastically improves the visual quality of the original images and outperforms the existing methods.



(a) Original **(b)** HE-Method **(c)** GA-Method

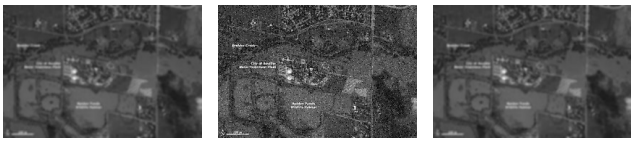
Fig. 4: Proposed Genetic Segmented Algorithm in image_0



(a) Original **(b)** HE-Method **(c)** GA-Method

Fig. 5: Proposed Genetic Segmented Algorithm in image_1

In addition to descriptive results, this manuscript contains a comparison with respect to other established measures, i.e., standard metrics like Mean, Variance, and Entropy. Table 4.1 and 4.2 gives quantitative results with respect to these methods. These tables contain numerical values associated with the images described before (Fig.6 - Fig.13). All these measures need to be taken care of as alteration of the histogram is distortion intensive task. Therefore it impacts all these measures. Keeping the image informative requires maintaining these parameters. Tables 4.1 and 4.2 show that all the image metrics taken from the above figures are within the permissible



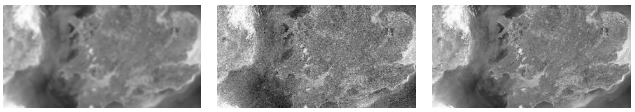
(a) Original **(b)** HE-Method **(c)** GA-Method

Fig. 6: Proposed Genetic Segmented Algorithm in image_2



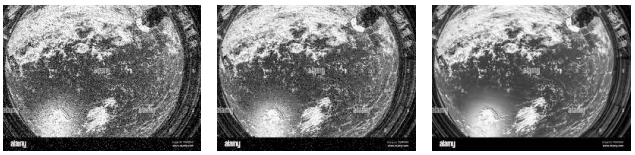
(a) Original **(b)** HE-Method **(c)** GA-Method

Fig. 7: Proposed Genetic Segmented Algorithm in image_3



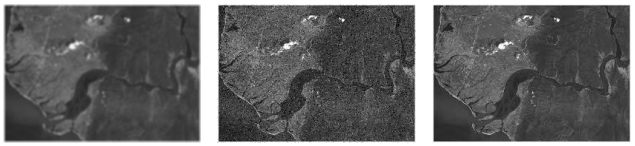
(a) Original **(b)** HE-Method **(c)** GA-Method

Fig. 8: Proposed Genetic Segmented Algorithm in image_4



(a) Original **(b)** HE-Method **(c)** GA-Method

Fig. 9: Proposed Genetic Segmented Algorithm in image_5



(a) Original **(b)** HE-Method **(c)** GA-Method

Fig. 10: Proposed Genetic Segmented Algorithm in image_6



(a) Original **(b)** HE-Method **(c)** GA-Method

Fig. 11: Proposed Genetic Segmented Algorithm in image_7

range[1] compared to other traditional histogram equalization-based techniques. While keeping the distortion low, the proposed can enhance the image's visual quality considerably. Therefore the proposed method is well suited for contrast enhancement problems.

Results shown in Table 4.1 and Table 4.2 consists of Mean Gray Values, Variance, and Entropy. If the gray values are distributed more widely, it implies it contains more information. The mean falls in the range of 115 - 140, concluding that the contrast of the pixels has been enhanced[1]. In comparison, the variance decides the deviation of values from the mean. Another measurable parameter is entropy, which is the logarithmic measure of an image's information. The higher the value of entropy, the more helpful information image contains. It can easily be observed from Table 4.2 that the entropy values of all the images have been improved in the proposed method as compared to the other existing methods.

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TABLE 4.1: Comparing Image Parameters of HE- Method and Proposed Genetic Segmented Algorithm

Images	HE-Method Mean	GA-Method Mean	HE-Method Variance	GA-Method Variance
image_0	119.041841	119.5439331	8.313466162	8.310727219
image_1	142.5	113.0132743	8.077124597	8.078161363
image_2	127.523622	127.0393701	8.566715646	8.559202732
image_3	133.4273859	120.4813278	8.391682428	8.342297794
image_4	143.4151786	111.9866071	8.044495126	8.043238694
image_5	127.9921569	127.4901961	8.579987917	8.583447048
image_6	0.5	155.5	0.707106781	6.204836823
image_7	127.5	126.5585938	8.703991245	8.688055662

TABLE II: Comparing Entropy and SSIM Parameters of HE-Method and Proposed GA segmented Method

Images	HE-Method Entropy	GA-Method Entropy	HE-Method SSIM	GA-Method SSIM
image_0	7.900866808	7.42387936	0.999999412	0.999998897
image_1	7.820178962	7.324603741	0.995992356	0.996269635
image_2	7.988684687	7.516243742	0.999999458	0.999998991
image_3	7.912889336	7.42326278	0.999612322	0.999878575
image_4	7.807354922	7.325212065	0.997719384	0.997762688
image_5	7.994353437	7.508078927	0.999999418	0.999999817
image_6	1	1	0.892831828	0.943361342
image_7	8	7.515625	0.999999404	0.999999892

Therefore it can be concluded that the proposed method is more effective and efficient as compared to state-of-the-art techniques.

V. Conclusion

In this paper, a genetic algorithm-based histogram modification method for image contrast enhancement has been proposed. The methodology works around modifying the original image using the segmented approach of the genetic algorithm. The segmented system lets us include flexibility to adjust the parameters for controlling the manual enhancement level for getting the new chromosomes and updating the value of the fitness function. It helps to preserve the image's finer details. This method has been evaluated against quantitative and qualitative parameters, i.e., Mean, Variance, Entropy, and SSIM. All these measures are essential to ensure no information loss in the modified histogram. In numeric terms, the mean is maintained between 115 - 140, which justifies the positive results because the mean is distorted and poorly maintained in the Histogram Equalization process. Therefore the proposed Genetic Algorithm based method has been designed, which carries forward the average pixel intensity value. The structural similarity value is close to 1 means the original image information has been retained in the enhanced image. Conclusively, the proposed method is very efficient concerning existing methods. Future work of this paper includes optimization of the proposed algorithm to reduce the complexity of running on low computational capability devices. Also, the current method can be further extended for the color images.

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